

LAB REPORT

Snort

Hands-On Network Intrusion Detection with Snort

IDS Fundamentals · Blue Team · Zen Mier

1. Objectives

This room takes a much deeper and more hands-on approach to Snort than the IDS Fundamentals lab. Rather than just learning what Snort is, this lab focuses on actually using it — running it against live traffic and recorded packet files, understanding its configuration in detail, writing rules, and interpreting the output it produces.

- Run Snort in sniffer, packet logger, and NIDS modes and understand when to use each.
- Analyze recorded network traffic (PCAP files) with Snort using both default and custom rule configurations.
- Understand the full structure of a Snort rule and write custom rules to detect specific traffic.
- Use Snort's traffic generator to practice detection against simulated attack scenarios.
- Count and interpret alerts generated from real packet captures.
- Apply Snort in IPS (blocking) mode to stop a live attack and retrieve the flag.

2. What Was Done

Running Snort and Checking Configuration

The first step was confirming the Snort installation and checking which rule sets were loaded. Running Snort with the default configuration file loaded 4,151 rules. A second configuration file (snortv2.conf) was also tested to compare. The build number and version details were retrieved using the -V flag.

```
ubuntu@ip-10-49-189-232:~$ snort -V
--
_*> Snort! <*_
o" )~ Version 2.9.7.0 GRE (Build 149)
' ' By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
Copyright (C) 2014 Cisco and/or its affiliates. All rights reserved.
Copyright (C) 1998-2013 Sourcefire, Inc., et al.
Using libpcap version 1.9.1 (with TPACKET_V3)
Using PCRE version: 8.39 2016-06-14
Using ZLIB version: 1.2.11

ubuntu@ip-10-49-189-232:~$ sudo snort -c /etc/snort/snort.conf -T
Running in Test mode

--== Initializing Snort ==--
Initializing Output Plugins!
Initializing Preprocessors!
Initializing Plug-ins!
Parsing Rules file "/etc/snort/snort.conf"
PortVar 'HTTP_PORTS' defined : [ 80:81 311 383 591 593 901 1220 1414 1741 1830 2301 2381 2809 3037 3128 3702 4343 4848 5250 6988 7000:7001
7144:7145 7510 7777 7779 8000 8008 8014 8028 8080 8085 8088 8090 8118 8123 8180:8181 8243 8280 8300 8800 8888 8899 9000 9060 9080 9090:909
1 9443 9999 11371 34443:34444 41080 50002 55555 ]
PortVar 'SHELLCODE_PORTS' defined : [ 0:79 81:65535 ]
PortVar 'ORACLE_PORTS' defined : [ 1024:65535 ]
PortVar 'SSH_PORTS' defined : [ 22 ]
PortVar 'FTP_PORTS' defined : [ 21 2100 3535 ]
PortVar 'SIP_PORTS' defined : [ 5060:5061 5600 ]
PortVar 'FILE_DATA_PORTS' defined : [ 80:81 110 143 311 383 591 593 901 1220 1414 1741 1830 2301 2381 2809 3037 3128 3702 4343 4848 5250 6
988 7000:7001 7144:7145 7510 7777 7779 8000 8008 8014 8028 8080 8085 8088 8090 8118 8123 8180:8181 8243 8280 8300 8800 8888 8899 9000 9060
9080 9090:9091 9443 9999 11371 34443:34444 41080 50002 55555 ]
PortVar 'GTP_PORTS' defined : [ 2123 2152 3386 ]
Detection:
  Search-Method = AC-Full-Q
  Split Any/Any group = enabled
  Search-Method-Optimizations = enabled
  Maximum pattern length = 20
```

Analyzing PCAP Files

Several PCAP files (mx-1.pcap, mx-2.pcap, mx-3.pcap) were analyzed using Snort with the default configuration. Running Snort against mx-1.pcap generated 170 alerts, with 18 TCP segments queued using the 1st configuration. Then the using the snortv2.conf use, it generate 68 alerts.

sudo snort -c /etc/snort/snort.conf -A full -l . -r mx-1.pcap

```
ubuntu@ip-10-49-189-232:~/Desktop/Task-Exercises/Exercise-Files/TASK-8$ sudo snort -c /etc/snort/snort.conf -A full -l . -r mx-1.pcap
Running in IDS mode

--== Initializing Snort ==--
Initializing Output Plugins!
Initializing Preprocessors!
Initializing Plug-ins!
Parsing Rules file "/etc/snort/snort.conf"
PortVar 'HTTP_PORTS' defined : [ 80:81 311 383 591 593 901 1220 1414 1741 1830 2301 2381 2809 3037 3128 3702 4343 4848 5250 6988 7000:7001
7144:7145 7510 7777 7779 8000 8008 8014 8028 8080 8085 8088 8090 8118 8123 8180:8181 8243 8280 8300 8800 8888 8899 9000 9060 9080 9090:909
1 9443 9999 11371 34443:34444 41080 50002 55555 ]
PortVar 'SHELLCODE_PORTS' defined : [ 0:79 81:65535 ]
PortVar 'ORACLE_PORTS' defined : [ 1024:65535 ]
PortVar 'SSH_PORTS' defined : [ 22 ]
PortVar 'FTP_PORTS' defined : [ 21 2100 3535 ]
PortVar 'SIP_PORTS' defined : [ 5060:5061 5600 ]

=====
Action Stats:
  Alerts:      170 (147.826%)
  Logged:      170 (147.826%)
  Passed:       0 ( 0.000%)
Limits:
  Match:       0
  Queue:       0
  Log:         0
  Event:       0
  Alert:       0

  TCP Segments Queued: 18
  TCP Segments Released: 18
  TCP Rebuilt Packets: 5
  TCP Segments Used: 18
  TCP Discards: 1
  TCP Gaps: 0
```

sudo snort -c /etc/snort/snortv2.conf -A full -l . -r mx-1.pcap

```
ubuntu@ip-10-49-189-232:~/Desktop/Task-Exercises/Exercise-Files/TASK-8$ sudo snort -c /etc/snort/snortv2.conf -A full -l . -r mx-1.pcap
Running in IDS mode

--== Initializing Snort ==--
Initializing Output Plugins!
Initializing Preprocessors!
Initializing Plug-ins!
Parsing Rules file "/etc/snort/snortv2.conf"
PortVar 'HTTP_PORTS' defined : [ 80:81 311 383 591 593 901 1220 1414 1741 1830 2301 2381 2809 3037 3128 3702 4343 4848 5250 6988 7000:7001
7144:7145 7510 7777 7779 8000 8008 8014 8028 8080 8085 8088 8090 8118 8123 8180:8181 8243 8280 8300 8800 8888 8899 9000 9060 9080 9090:909
1 9443 9999 11371 34443:34444 41080 50002 55555 ]
PortVar 'SHELLCODE_PORTS' defined : [ 0:79 81:65535 ]
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PortVar 'SSH_PORTS' defined : [ 22 ]
PortVar 'FTP_PORTS' defined : [ 21 2100 3535 ]
PortVar 'SIP_PORTS' defined : [ 5060:5061 5600 ]
PortVar 'FILE DATA PORTS' defined : [ 80:81 110 143 311 383 591 593 901 1220 1414 1741 1830 2301 2381 2809 3037 3128 3702 4343 4848 5250 6
988 7000:7001 7144:7145 7510 7777 7779 8000 8008 8014 8028 8080 8085 8088 8090 8118 8123 8180:8181 8243 8280 8300 8800 8888 8899 9000 9060
9080 9090:9091 9443 9999 11371 34443:34444 41080 50002 55555 ]
PortVar 'GTP_PORTS' defined : [ 2123 2152 3386 ]
Detection:
  Search-Method = AC-Full-Q

=====
Action Stats:
  Alerts:      68 ( 59.130%)
  Logged:      68 ( 59.130%)
  Passed:       0 ( 0.000%)
```

mx2.pcap and mx3.pcap were combined in a single run to produce a combined alert count of 1,020. The output logs were stored in the current directory and reviewed to confirm the alert types triggered.

sudo snort -c /etc/snort/snort.conf -A full -l . --pcap-list="mx-2.pcap mx-3.pcap"

```
ubuntu@ip-10-49-189-232:~/Desktop/Task-Exercises/Exercise-Files/TASK-8$ sudo snort -c /etc/snort/snort.conf -A full -l . --pcap-list="mx-2.
pcap mx-3.pcap"
Running in IDS mode

--== Initializing Snort ==--
Initializing Output Plugins!
Initializing Preprocessors!
Initializing Plug-ins!
Parsing Rules file "/etc/snort/snort.conf"
PortVar 'HTTP_PORTS' defined : [ 80:81 311 383 591 593 901 1220 1414 1741 1830 2301 2381 2809 3037 3128 3702 4343 4848 5250 6988 7000:7001
7144:7145 7510 7777 7779 8000 8008 8014 8028 8080 8085 8088 8090 8118 8123 8180:8181 8243 8280 8300 8800 8888 8899 9000 9060 9080 9090:909
1 9443 9999 11371 34443:34444 41080 50002 55555 ]
PortVar 'SHELLCODE_PORTS' defined : [ 0:79 81:65535 ]
PortVar 'ORACLE_PORTS' defined : [ 1024:65535 ]
PortVar 'SSH_PORTS' defined : [ 22 ]
PortVar 'FTP_PORTS' defined : [ 21 2100 3535 ]
PortVar 'SIP_PORTS' defined : [ 5060:5061 5600 ]
```

```
=====
Action Stats:
  Alerts:      1020 (147.826%)
  Logged:      1020 (147.826%)
  Passed:       0 ( 0.000%)
```

Writing Custom Detection Rules

Custom rules were written in local.rules to detect specific traffic patterns. A rule was created to detect all packets with Syn and Push-Ack flags, and UDP packets in task9.pcap, and the number of matching packets was counted from the output. Rules follow the format: action, protocol, source, direction arrow, destination, followed by rule options in brackets including msg (alert message), sid (unique rule ID), and rev (revision number).

```
GNU nano 4.8                                local.rules                                Modified
# -----
# LOCAL RULES
# -----
# This file intentionally does not come with signatures.  Put your local
# Filter packets with Syn flags
alert tcp any any <> any any (msg:"SYN Flag Filter"; flags:S; sid:1000002; rev:1;)
# Filter packets with Push-Ack flags
alert tcp any any <> any any (msg:"Push-Ack Filter"; flags:PA; sid:1000003; rev:1;)
# Filter UDP packets
alert udp any any <> any any (msg:"Same Source and Destination IP"; sameip; sid:1000004; rev:1;)

ubuntu@ip-10-49-189-232:~/Desktop/Task-Exercises/Exercise-Files/TASK-9$ snort -c local.rules -A full -l . -r task9.pcap
Running in IDS mode

--== Initializing Snort ==--
Initializing Output Plugins!
Initializing Preprocessors!
Initializing Plug-ins!
Parsing Rules file "local.rules"
Tagged Packet Limit: 256
Log directory = .

+++++
Initializing rule chains...
1 Snort rules read
  1 detection rules
  0 decoder rules
  0 preprocessor rules
1 Option Chains linked into 1 Chain Headers
0 Dynamic rules
```

3. Summary

This lab moved from theory into practical, applied Snort usage. The ability to write a rule, run it against real or live traffic, and interpret the output is a fundamental skill for any network defender. The IPS exercises especially bridged the gap between detection and active response — turning Snort from a monitoring tool into a blocking tool.

Area Covered	Key Takeaway
PCAP Analysis	Snort generated 170 alerts from mx-1.pcap and 1,020 alerts from a combined mx2.pcap and mx3.pcap run.
Custom Rules	Written in local.rules — detected HTTP, SSH, ICMP, and other protocol traffic by port and pattern.
Rule Structure	Action + Protocol + Source/Dest IPs & Ports + Options (msg, sid, rev).
IPS Mode	Snort configured to block a live brute-force SSH attack and a reverse shell connection.
Configuration	Default config loaded 4,151 rules; custom rules extend detection beyond the built-in ruleset.